HARNESSING IOT FOR EFFICIENT IRRIGATION

A Project Report submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

INFORMATION TECHNOLOGY

BY:
MANYA MEENA(2022UIT3142)

Under the supervision of
DR. NISHA KANDHOUL

Information Technology (IT)
Netaji Subhas University of Technology, Delhi

DIVISION OF INFORMATION TECHNOLOGY
NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY
DELHI-110078
May 22, 2026

CERTIFICATE

This is to certify that the project titled **HARNESSING IOT FOR EFFICIENT IRRIGATION** is a bonafide record of the work submitted by **MANYA MEENA**(2022UIT3142) under my supervision and guidance in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology of the **NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY, DELHI-110078** during the year 2025.

"To the best of my knowledge, their work is original and has not been submitted for any other degree or qualification."

DATE:15-04-2025

DR. NISHA KANDHOUL
PROFESSOR
DEPARTMENT OF INFORMATION TECHNOLOGY
NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

DECLARATION

This is to certify that the project entitled "**HARNESSING IOT FOR EFFICIENT IRRIGATION SYSTEM**" submitted by the undersigned in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering is a bonafide and original work carried out under the supervision of Dr. **Nisha Kandhoul**, Department of Information Technology, Netaji Subhas University of Technology, University of Delhi, New Delhi, during the period January 2025 to April 2025.

To the best of our knowledge and belief, the content of this project report represents the candidates' original work and has not been submitted, either in part or in full, for the award of any other degree or diploma at this or any other degree elsewhere.

MANYA MEENA(2022UIT3142)
DATE:15-04-2025

ACKNOWLEDGEMENT

We take this opportunity to sincerely acknowledge the contributions and support of all those who made the successful completion of this project possible.

First, we express our heartfelt gratitude to our project supervisor, **Dr. Nisha Kandhoul**, Department of Information Technology, Netaji Subhas University of Technology, for her invaluable guidance, unwavering support and constructive feedback throughout the duration of this project. Her expertise, encouragement, and insightful suggestions were instrumental in shaping the direction and quality of our work.

We are also grateful to the esteemed faculty members of the Department of Information Technology, Netaji Subhas University of Technology, for imparting the foundational knowledge and providing access to essential resources and infrastructure.

Lastly, we extend our deepest appreciation to our families and friends for their continued encouragement, patience, and support throughout our academic journey.

TABLE OF CONTENTS

List of Figures 5

1 Introduction 1

2 AIM AND SCOPE OF THE PRESENT INVESTIGATION 4

3 System Integration and Components 7

4 SOFTWARE 14

5 RESULTS AND PERFORMANCE ANALYSIS 17

6 SUMMARY AND CONCLUSIONS 20

Bibliography 21

List of Figures

1.1 Steps performed in smart agriculture system 3

2.1 System prototype 6

3.1 NodeMCU ESP8266 9
3.2 Soil moisture sensor 10
3.3 Jumper wires 11
3.4 Relay module 12
3.5 Water pump 13

4.1 Blynk dashboard 15

5.1 Configuration of NodeMCU Board 17
5.2 Determines soil moisture status 18
5.3 Soil status and moisture level 18
5.4 Device online status 19

Chapter 1

Introduction

Soil moisture information plays an important role in environmental monitoring, agricultural production and hydrological studies. Particularly, agricultural yield depends on several growing parameters like temperature, humidity, soil moisture and pH of the soil, etc. In this project, we have designed and developed a system for measuring and monitoring soil moisture by interfacing low-cost soil moisture sensor with Internet of Things (IoT), Cloud computing and Mobile computing technologies.

1.1 INTERNET OF THINGS

The Internet of Things (IoT) refers to a network of interconnected physical devices embedded with sensors, software, and other technologies that enable them to collect and exchange data over the Internet or other communication networks. These devices, often referred to as "smart" devices or "connected" devices, can include everything from home appliances and wearable health monitors to industrial machines and agricultural sensors.

The primary objective of IoT is to enable real-time monitoring, automation, and control of systems without human intervention. Devices in an IoT ecosystem communicate with each other and with centralized platforms or cloud services to share data, analyze it, and trigger intelligent actions based on predefined conditions or machine learning algorithms.

Applications of IoT:

- **Smart Agriculture:** Automated irrigation, soil monitoring, crop health analysis.
- **Smart Homes:** Lighting, temperature control, security systems.
- **Healthcare:** Remote patient monitoring, wearable health trackers.
- **Industrial IoT (IIoT):** Predictive maintenance, process automation.
- **Smart Cities:** Traffic control, waste management, energy monitoring.

1.2 OVERVIEW OF THE PROJECT

In India, agriculture contributes about 16 Percent of total GDP and 10 percent of total exports. Water is main resource for Agriculture. Irrigation is one method to supply water but in some cases, there will be lot of water wastage. Therefore, in this regard to save water and time we have proposed project titled NodeMCU ESP8266 Wi-Fi module based automatic irrigation system using Arduino IDE-IoT. In this proposed system, we are using various sensors like temperature, humidity, soil moisture sensors that sense the various parameters of the soil. In addition, based on soil moisture value land is automatically irrigated by ON/OFF of the motor. These sensed parameters and motor status will be displayed on user android application. The Internet of Things (IoT) is a technology where a mobile device can be used to monitor the function of a device. The Internet of Things (IoT) is concerned with interconnecting communicating objects that are installed at different locations that are possibly distant from each other. Internet of Things (IoT) is a type of network technology, which senses the information from different sensors and make anything to join the Internet to exchange information. It can also be used to modify the status of the device. The central processing unit will also include communication device to receive data from the sensors and to be relayed to the user's device. This will be done using a higher communication device such as a Wi-Fi module. The data processed by the central module is converted to meaningful data and relayed to the user. The user can view the data with the help of a handheld device such as a mobile phone or a tablet. Nowadays water scarcity is a big concern for farming. This project helps the farmers to irrigate the farmland in an efficient manner with automated irrigation system based on soil moisture.

1.2.1 RELATED WORKS

There are many techniques to perform this soil moisture sensing using not only ESP8266, we can also use Arduino board and many other cloud platforms such as Thingspeak Cloud, AWS Cloud, adafruit etc., we can connect to many cloud databases. Nowadays, for irrigation, different techniques are available which are used to reduce the dependency of rain. And mostly this technique is driven by electrical power and on/off scheduling. In this technique, soil moisture sensors are placed root zone of plant and near the module and gateway unit handles the sensor information.

1.3 EXISTING SYSTEM

Many existing, well established soil moisture systems are based on wired communication. IoT is a system that uses computers or mobile devices to control basic agriculture functions and features automatically through Internet from anywhere around the world. Proposed system is using free IoT platform, ArduinoIDE using NodeMCU Wi-Fi web service to configure the smart agriculture. It is very simple and user free smart agriculture system compared to existing system.

1.4 BLOCK DIAGRAM

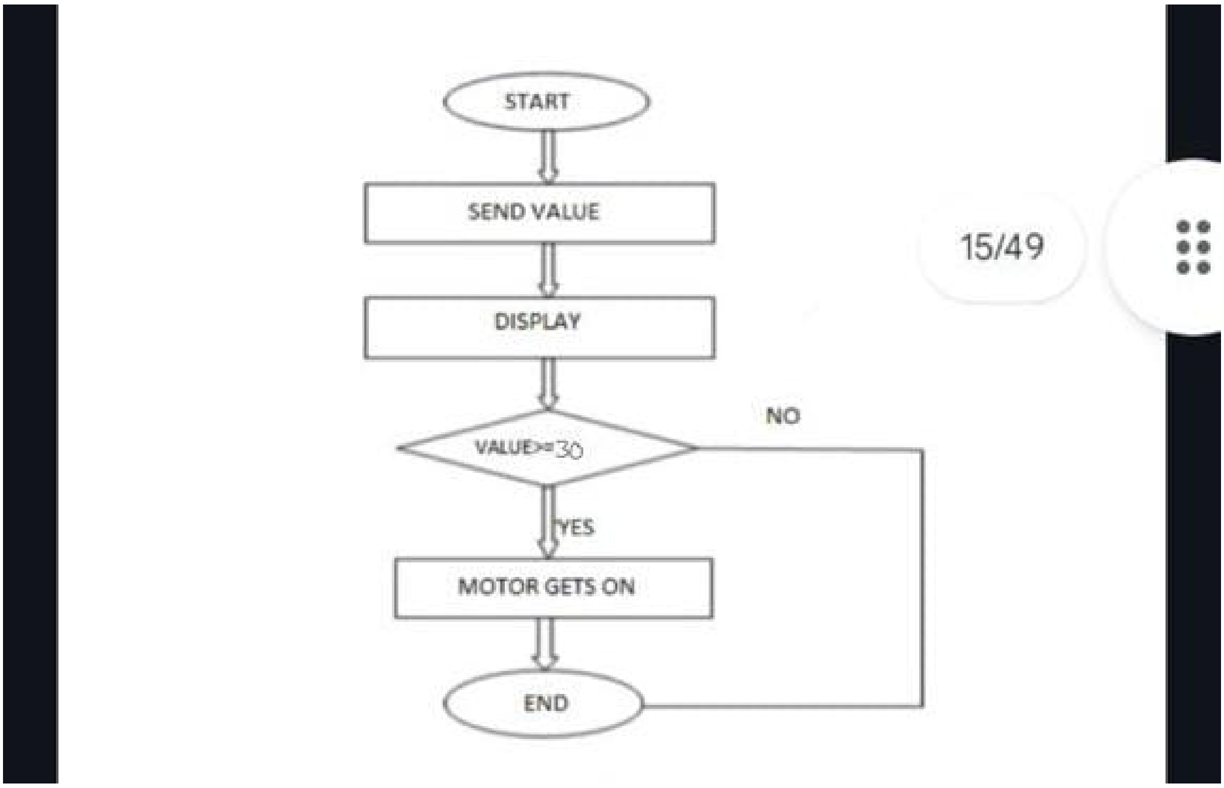

Figure 1.1: Steps performed in smart agriculture system

Chapter 2

AIM AND SCOPE OF THE PRESENT INVESTIGATION

The proposed system has been designed to overcome the unnecessary water flow into the agricultural lands. Moisture readings are continuously monitored by using moisture sensor and send these values to the assigned IP address. Android application continuously collects the data from that assigned IP address. Once the soil moisture values are exceeded the particular limit then the relay, which is connected to the NodeMCU microcontroller controls the motor. The android application is a simple menu driven application. This includes motor status, moisture values. The motor status indicates the current status of the pump.

2.1 OBJECTIVE OF THE PROJECT

The main objective of this project is to provide an automatic irrigation system thereby saving time, money and power of the farmer. The traditional farm-land irrigation techniques require manual intervention with the automated technology of irrigation the human intervention can be minimized. Most of the farmers use large portions of farming land and it becomes very difficult to reach and track each corner of large lands. Sometime there is a possibility of uneven water sprinkles or less moisture content . This result in the bad quality crops which further leads to financial losses. In this scenario the Smart Irrigation System using Latest IoT technology is helpful and leads to ease of farming. The Smart Irrigation System has wide scope to automate the complete irrigation system. Here we are building a IoT based Irrigation System using ESP8266 NodeMCU Module and moisture Sensor. It will not only automatically irrigate the water based on the moisture level in the soil but also send the Data to IoT Cloud Server to keep track of the land condition. The System will consist a water pump which will be used to sprinkle water on the land depending upon the land environmental condition such as Moisture. Excessive water use for agriculture is leaving rivers, lakes and underground water sources dry in many irrigated areas, according to the World Wildlife Fund. The agricultural sector uses 70 percent of the world's accessible fresh water, which is three times more than industry and more than eight times more than municipalities use. So with this procedure using soil moisture with thinkspeak Cloud we can save water and preserve it for future use.

2.2 ADVANTAGES OF PROJECT

- Simple method of measurement.
- It delivers the results immediately.

- Watermark sensors are very low in cost.
- Offers accurate results.
- Watermark sensors offer larger moisture reading range.

2.3 DISADVANTAGES OF PROJECT

- It requires initial evaluation of site specific conditions before selection of appropriate moisture sensor.
- It requires probe to be inserted in the soil. It requires labor to collect the data and maintain the measurement processes.
- The measured values depend on properties of various materials. The correct interpretation and use of moisture data is needed.
- Watermark sensors provide less accuracy in sandy soils due to large particles.
- Watermark sensors are required to be calibrated for each soil types.

2.4 SCOPE

The proposed agricultural system is designed to solve to find an optimal solution to the water crisis. The design implements IoT technology using an android device, a main controlling unit (NodeMCU), sensors to measure various parameters and a water pump, which will be used to supply water to the farm.

2.4.1 PROTOTYPE

The proposed system is designed for automated irrigation based on real-time soil moisture monitoring, with the primary objective of optimizing water usage and promoting efficient agricultural practices. The system is centered around the NodeMCU, an open-source IoT platform based on the ESP8266 Wi-Fi module, which serves as the main controller and communication unit.

A soil moisture sensor is interfaced with the NodeMCU to continuously monitor the water content present in the soil. The sensor operates by measuring the dielectric permittivity of the soil, which changes with the moisture level. The sensor outputs an analog or digital signal proportional to the moisture content, which is read and processed by the NodeMCU.

When the moisture level falls below a predefined threshold, the NodeMCU activates a relay module, which acts as an electronic switch. The relay, in turn, controls a water pump or motor, enabling the system to deliver water to the soil automatically. Once the moisture level reaches the desired limit, the motor is turned off, thereby preventing over-irrigation.

The relay module serves as an interface between the low-power microcontroller and the high-voltage motor circuit, ensuring safe and isolated switching operations. The use of a relay provides reliable control of high-current devices like motors while maintaining the safety and integrity of the low-voltage NodeMCU circuit.

In addition to basic automation, the NodeMCU's built-in Wi-Fi capability allows for remote monitoring and control. The data collected from the sensor can be transmitted to a cloud platform or a mobile application, enabling users to track soil conditions in real time and adjust irrigation settings as needed.

This system is particularly useful for smart farming applications, where precision and resource management are critical. It reduces manual intervention, conserves water, improves crop yield, and paves the way for scalable IoT-based agricultural solutions.

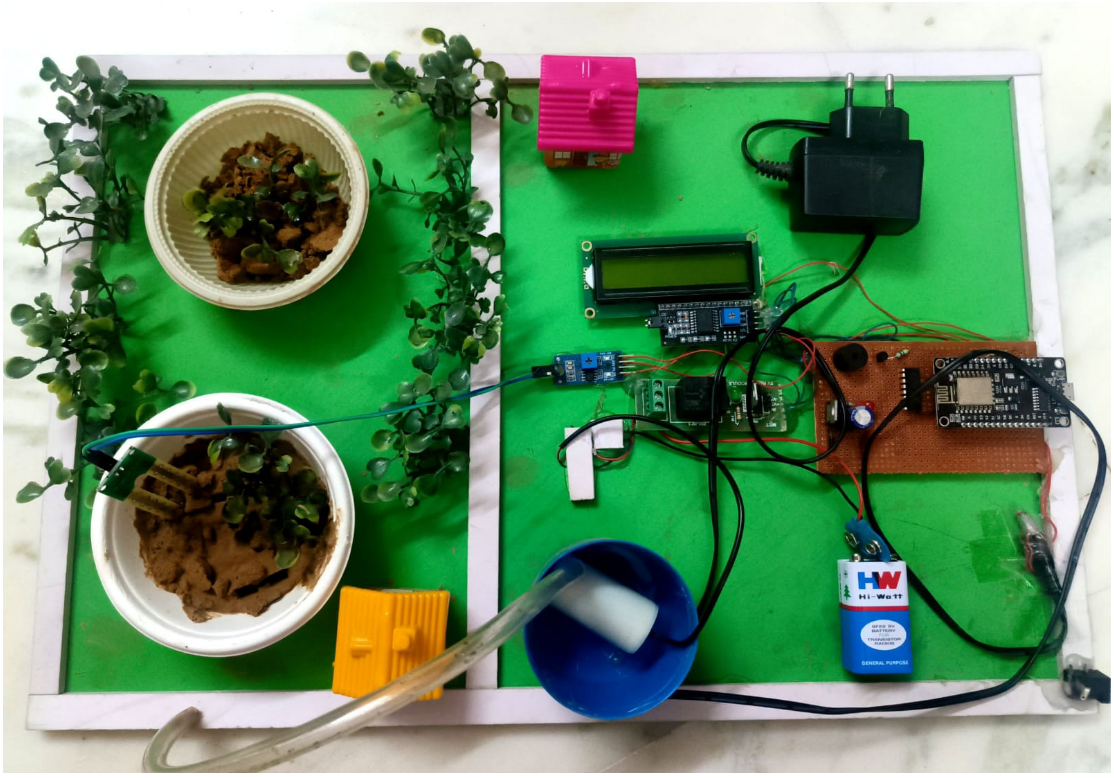

Figure 2.1: System prototype

Chapter 3

System Integration and Components

The Smart Agriculture System has wide scope to automate the complete irrigation system. Here we are building a IoT based Irrigation System using ESP8266 NodeMCU Module. It will not only automatically irrigate the water based on the moisture level in the soil but also send the Data to blynk app to keep track of the moisture in the soil. The System will consist a motor connected to it which will be used to spray water on the land depending upon the land environmental condition such as Moisture. Before starting, it is important to note that the different crops require different Soil Moisture condition. So, if a soil has a moisture of below 30. So, then Motor will turn on automatically to sprinkle the water and it will continue to sprinkle the water until the moisture goes up to above 30 and after that the pump will be turned off. The sensor data will be sent to blynk app in defined interval of time so that it can be monitored from anywhere in the world.

3.1 COMPONENTS REQUIRED

- NodeMCU ESP8266
- Soil Moisture Sensor Module
- Jumper wires
- Relay module
- Battery
- Water pump
- LCD Display

3.1.1 NODEMCU

NodeMCU is an open-source IoT development board and firmware based on the ESP8266 Wi-Fi module, designed to enable easy prototyping of IoT applications. The name "NodeMCU" stands for Node MicroController Unit, and it integrates both hardware and software components for developing connected devices.

3.1.1.1 Key Features:

- **ESP8266 Microcontroller:** A low-cost, power-efficient microchip with built-in Wi-Fi capability, serving as the core of the NodeMCU.

- **Built-in Wi-Fi:** Enables seamless connection to the internet or local networks without the need for additional hardware.
- **GPIO Pins:** General Purpose Input/Output pins allow connection to external sensors, actuators, and other modules.
- **USB Interface:** Easily programmable via a micro-USB connection using standard serial communication.
- **Programming Support:** Can be programmed using Lua scripting language or more commonly via the Arduino IDE using C/C++.

3.1.1.2 Technical Specifications:

- **Processor:** 32-bit RISC CPU, running at 80 MHz (can be overclocked to 160 MHz)
- **Memory:** 64 KB instruction RAM, 96 KB data RAM
- **Flash Memory:** Typically 4 MB (varies by version)
- **Operating Voltage:** 3.3V (with onboard voltage regulator for 5V input via USB)
- **Connectivity:** IEEE 802.11 b/g/n Wi-Fi

3.1.1.3 Advantages:

- Compact and lightweight
- Low power consumption
- Cost-effective
- Supports OTA (Over-the-Air) updates
- Rich community support and extensive libraries

3.1.1.4 Applications:

- Smart home automation systems
- Weather stations
- Smart agriculture (e.g., automated irrigation)
- Remote monitoring systems
- Wireless sensor networks

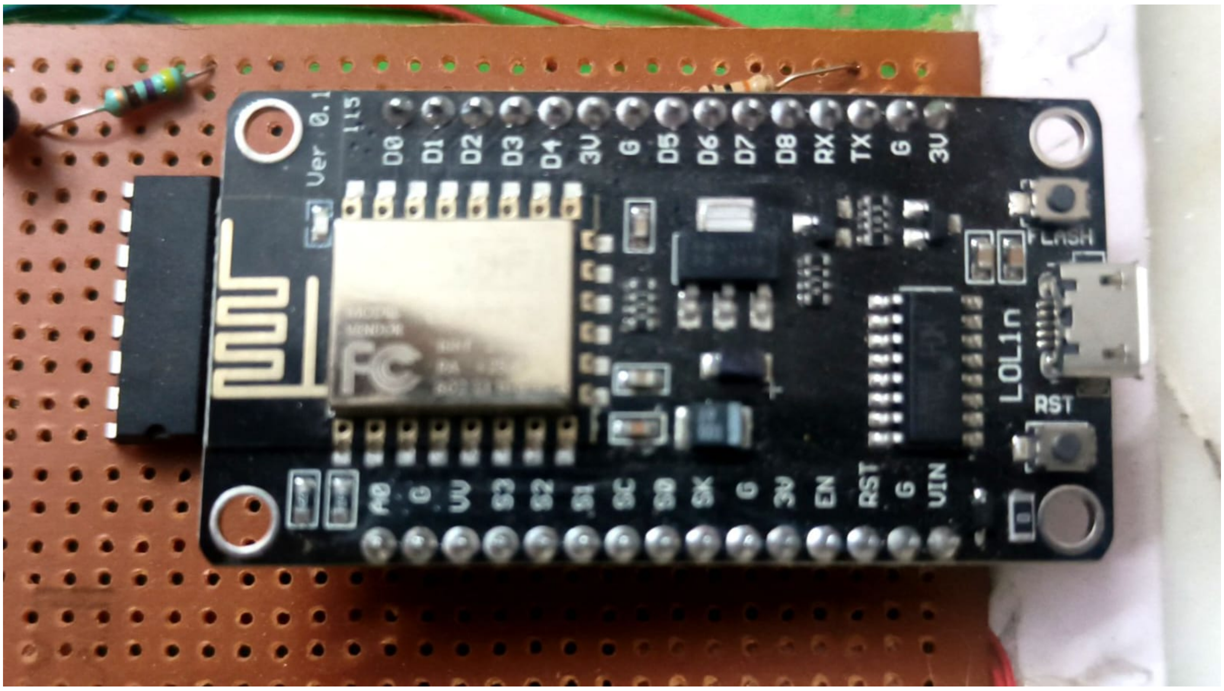

Figure 3.1: NodeMCU ESP8266

3.1.2 SOIL MOISTURE SENSOR

Soil Moisture Sensor

A Soil Moisture Sensor is an electronic device used to measure the volumetric water content in the soil. It plays a critical role in smart irrigation systems and precision agriculture, allowing real-time monitoring of soil conditions to optimize water usage and improve crop yield.

Working Principle:

The sensor typically consists of two probes that are inserted into the soil. It measures the electrical resistance or capacitance between the probes: Dry soil has high resistance (low conductivity), Wet soil has low resistance (high conductivity).

This analog or digital signal is then processed by a microcontroller (e.g., NodeMCU) to determine whether the soil needs watering.

Types of Soil Moisture Sensors:

- **Resistive Sensors** - Measure resistance; low cost, suitable for basic applications.
- **Capacitive Sensors** - Measure the dielectric permittivity of the soil; more accurate and durable.
- **Tensiometers** - Measure the tension with which water is held in the soil; used in research and high-precision farming

3.1.3 JUMPER WIRES

Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed. A jump wire (also known as jumper wire, or jumper) is an electrical wire, or group of them in a cable, with a connector or pin at each end (or sometimes without them - simply "tinned"), which is normally used to interconnect the components

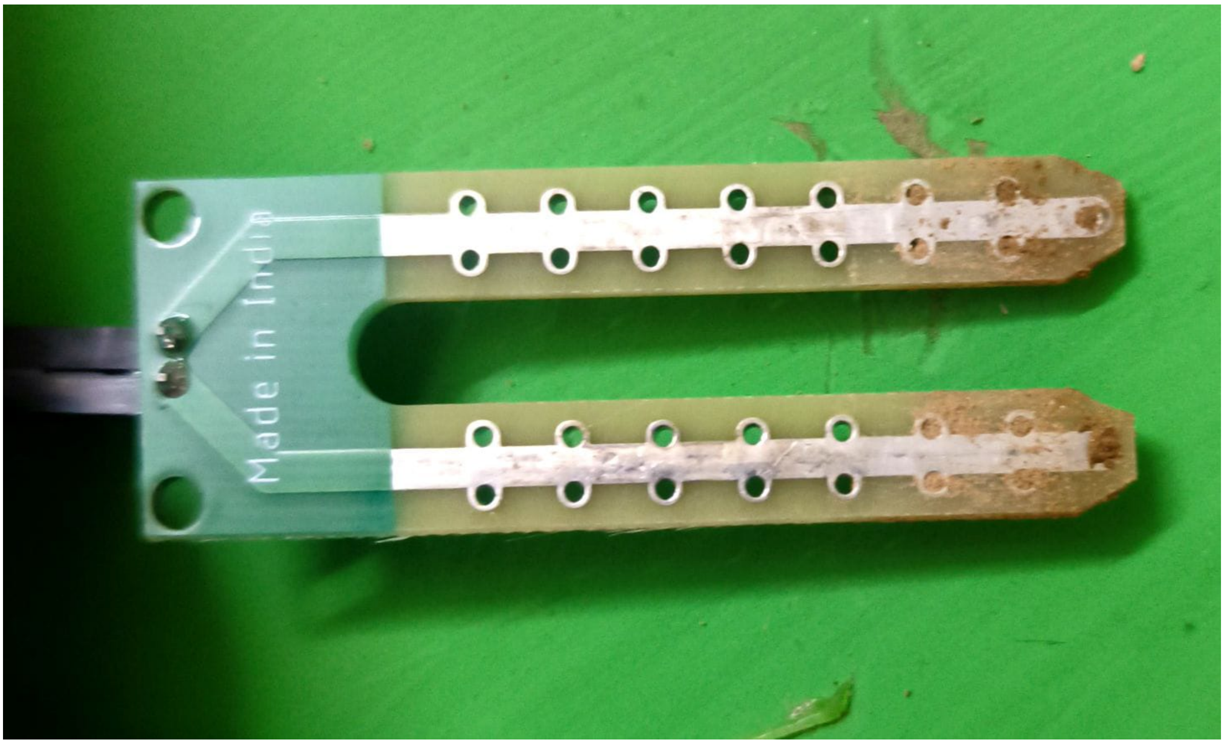

Figure 3.2: Soil moisture sensor

of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering. Individual jump wires are fitted by inserting their "end connectors" into the slots provided in a breadboard, the header connector of a circuit board, or a piece of test equipment.

3.1.4 RELAY MODULE

A Relay Module is an electrically operated switch used to control high-voltage or high-current devices—such as motors, lights, or pumps—using low-power signals from microcontrollers like NodeMCU, Arduino, or Raspberry Pi. It acts as an interface between the low-voltage control circuitry and high-power electrical components, providing electrical isolation and safe switching.

Working Principle:

A relay operates based on electromagnetic induction. When a small voltage is applied to the control coil, it generates a magnetic field that moves an internal switch, either opening or closing the circuit connected to a higher voltage source. Most commonly used relay modules in IoT and embedded systems are electromechanical relays, though solid-state relays (SSRs) are also available for faster switching and longer life.

Key Components:

- **Electromagnetic Relay:** The main switching device.
- **Transistor/Driver Circuit**: Amplifies the control signal from the microcontroller.
- **Flyback Diode:** Protects the circuit from voltage spikes caused by coil de-energization.
- **Optocoupler** (optional): Provides additional electrical isolation between the control and power sides.

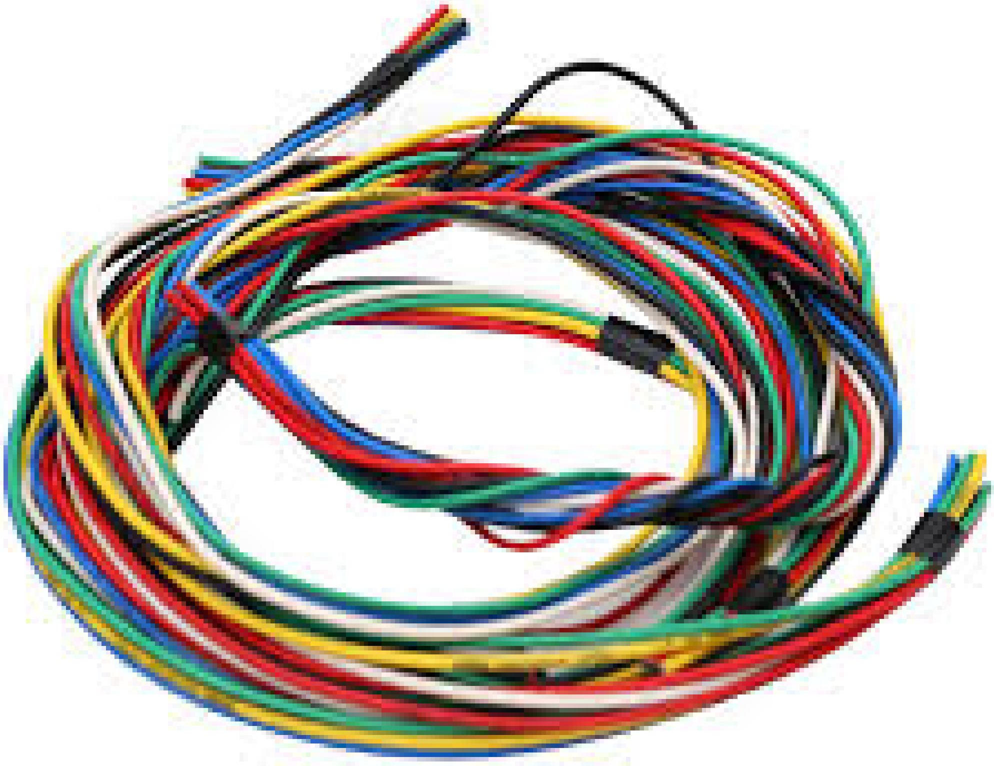

Figure 3.3: Jumper wires

- **Indicator LED:** Shows relay status (on/off).

3.1.5 WATER PUMP

Water Pump

A Water Pump is an electromechanical device used to move water from one location to another. In IoT-based smart irrigation systems, water pumps are commonly employed to deliver water to agricultural fields based on real-time soil moisture levels, enabling efficient and automated water management.

Types of Water Pumps Used in IoT Projects:

Submersible Pump - Operates while submerged in water; commonly used in wells or water tanks. DC Mini Pump - Compact and low-powered; suitable for small-scale irrigation setups. Centrifugal Pump - Uses rotational energy to move water; suitable for larger flow requirements.

Working Principle:

A water pump converts electrical energy into mechanical energy, typically through a rotating impeller or diaphragm mechanism. When powered, it creates a pressure difference, causing water to flow through the system.

In automated systems, the pump is typically connected to a relay module, which acts as a switch controlled by a microcontroller (such as NodeMCU). The microcontroller activates the pump when the soil moisture sensor detects dryness and turns it off once adequate moisture is reached.

Specifications (for small DC water pumps):

- Operating Voltage: 3V - 12V DC (commonly 5V or 6V in IoT setups)

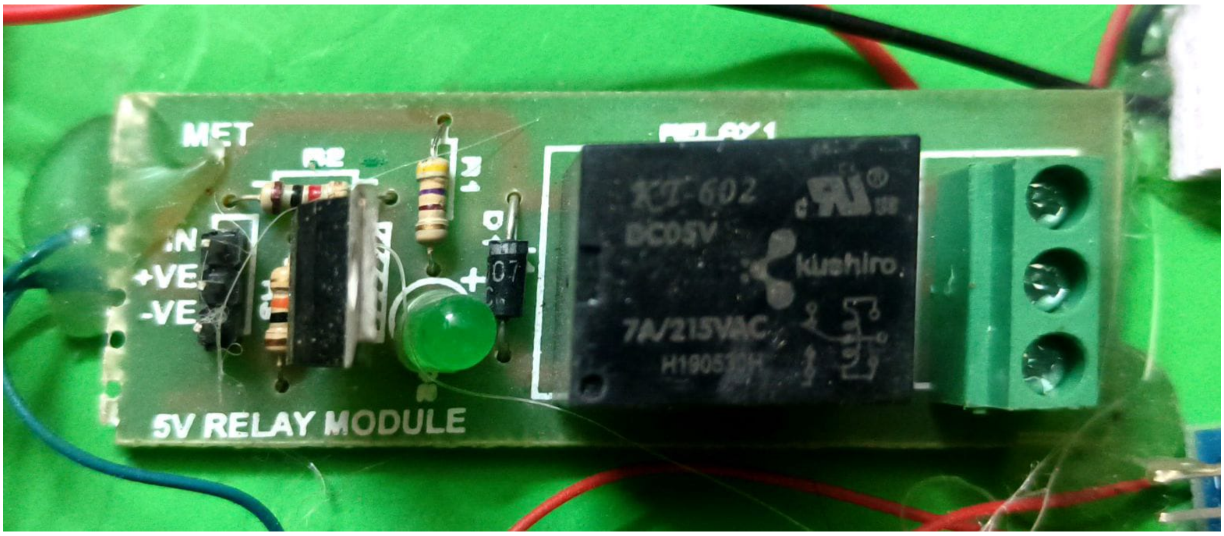

Figure 3.4: Relay module

- Flow Rate: 80 - 300 L/h (liters per hour)
- Current Draw: 100 mA - 500 mA
- Lift Distance: 0.5 to 2 meters
- Inlet/Outlet Diameter: 6mm - 8mm (typical)

Applications:

- Smart irrigation systems
- Hydroponics and greenhouse watering
- Garden automation
- Aquariums and water circulation systems

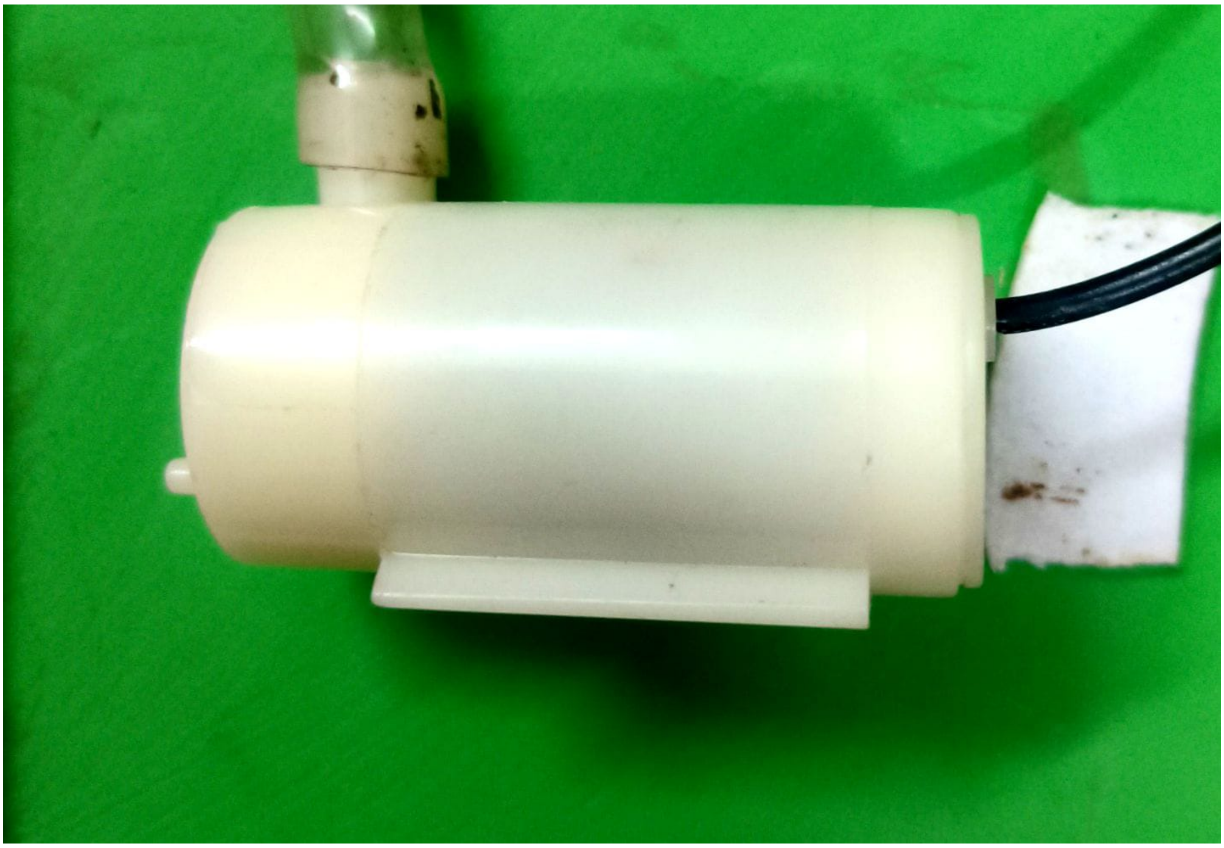

Figure 3.5: Water pump

Chapter 4

SOFTWARE

4 BLYNK APPLICATION

4.1.1 Blynk Application

Blynk is a platform for building IoT applications that enables users to control hardware remotely, display sensor data, store data, and perform automation using a smartphone or tablet. It provides a powerful and user-friendly interface for developing connected solutions without the need for complex frontend programming.

Initially released in 2015, Blynk supports a wide range of microcontrollers and development boards like NodeMCU (ESP8266/ESP32), Arduino, Raspberry Pi, and more. It allows communication between hardware devices and mobile applications using the Blynk Cloud, or a private server in offline/secure environments.

Key Features of Blynk:

- Drag-and-Drop Interface: Allows users to build mobile dashboards using widgets (buttons, sliders, displays, graphs, etc.).
- Real-Time Monitoring: View sensor data (e.g., temperature, moisture, humidity) in real-time.
- Device Control: Switch devices (e.g., motors, lights) on/off from the app.
- Data Logging: Store data in the cloud for later analysis.
- Notifications: Receive push notifications and alerts based on sensor values or events.
- Multi-Device Support: Manage and control multiple IoT devices within one app.

4.1.2 Core Components of Blynk:

- Blynk App (Mobile Application)-

Used to create user interfaces (dashboards) by placing widgets to interact with the connected hardware.

- Blynk Library-

Installed in the Arduino IDE or other development environments, this library allows the microcontroller to communicate with the Blynk Cloud or local server.

- Blynk Cloud-

A cloud service that handles communication between the hardware and the mobile app. Users can also set up a local Blynk server for offline or private use.

- Authentication Token-

Each project in Blynk generates a unique Auth Token which must be added to the code on the hardware to link it with the app. Cross-Platform: Available on both iOS and Android devices.

4.1.3 Typical Use Case in Smart Irrigation:

In an automated irrigation system:

The soil moisture sensor reads data and sends it to the NodeMCU. NodeMCU sends this data to the Blynk Cloud via Wi-Fi. The Blynk App displays real-time moisture levels. The user can manually trigger the water pump through a button widget or allow automatic control using code logic in the microcontroller. Users can receive notifications when moisture drops below a threshold.

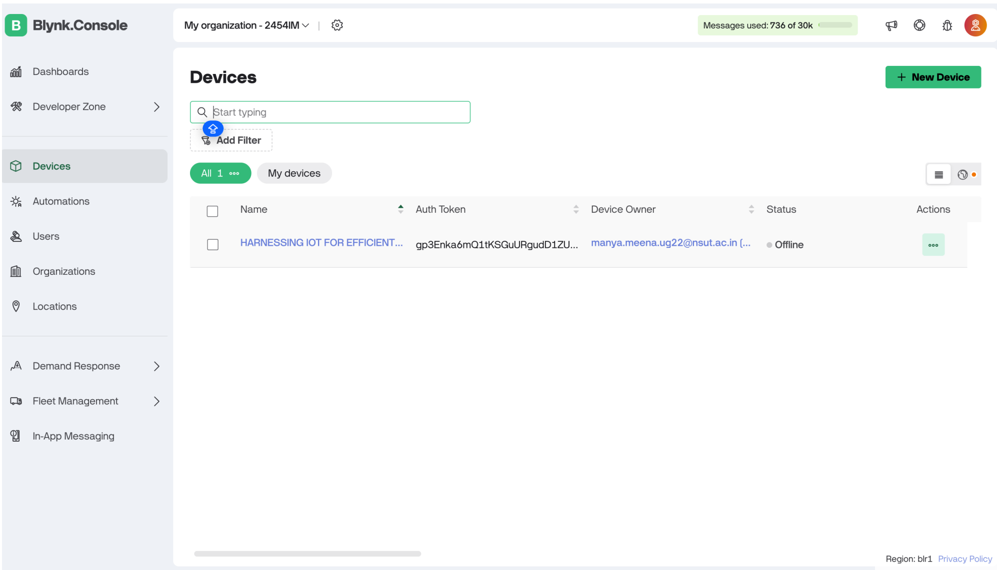

Figure 4.1: Blynk dashboard

4.2 ARDUINO-IDE

Arduino IDE (Integrated Development Environment)

The Arduino IDE is a free, open-source software platform used to write, compile, and upload code to Arduino-compatible development boards, such as Arduino Uno, NodeMCU, ESP8266, ESP32, and others. It is a critical tool for programming microcontrollers in IoT, embedded systems, robotics, and automation projects. Designed with simplicity in mind, the Arduino IDE makes embedded programming accessible to both beginners and professionals. It supports a C/C++-based programming language, along with a large number of built-in libraries and third-party extensions.

Key Features:

- Code Editor: Simple and clean interface to write and edit sketches (Arduino programs).

- Compiler: Translates Arduino code into machine code.
- Serial Monitor: Provides real-time data communication between the board and the computer, useful for debugging.
- Board Manager: Allows installation of support packages for various development boards (e.g., ESP8266, ESP32).
- Library Manager: Facilitates easy inclusion of external libraries for sensors, actuators, displays, Wi-Fi modules, etc.
- Cross-Platform: Available on Windows, macOS, and Linux.

Chapter 5

RESULTS AND PERFORMANCE ANALYSIS

5.1 Initialization of board

Initially import the board data from tools-> board-> NodeMCU 1.0.

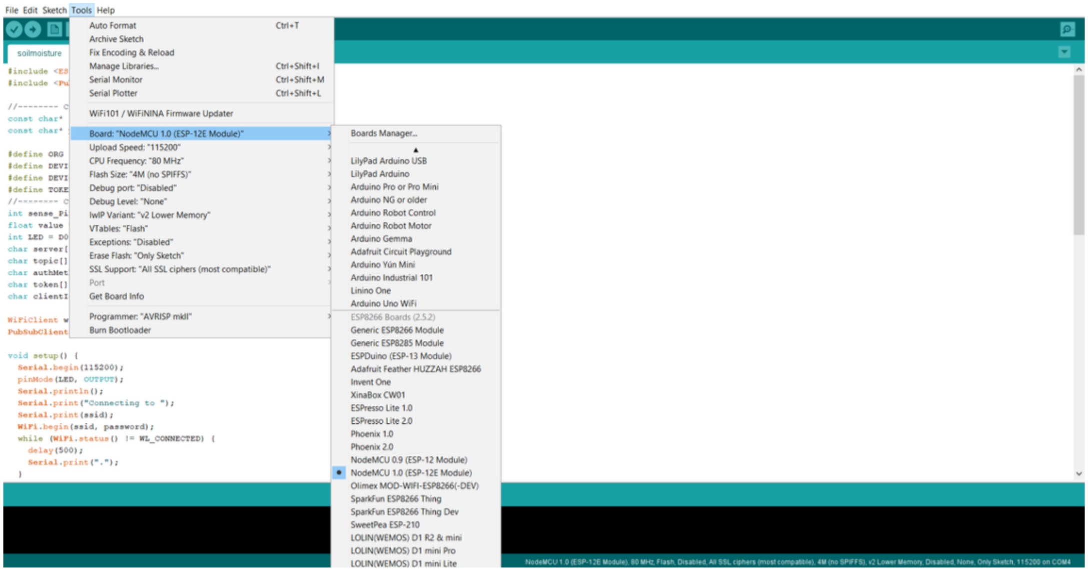

Figure 5.1: Configuration of NodeMCU Board

Compile and upload the code to the board.
Open the serial monitor, readings will update continuously with a delay of 2sec each.

This block executes when the soil moisture level falls below 30 percent, indicating dry soil conditions.
The status variable is updated to "Low" for potential display or logging.
The relay is triggered by setting it to HIGH, which activates the water pump to irrigate the soil.
Simultaneously, the buzzer is turned on as an alert signal to indicate low moisture.
IF 60 Percent or above, indicating sufficiently wet soil.
The status is set to "High".

```
// Determine Soil Moisture Status
if (moisturePercent < 30) {
    status = "Low";
    digitalWrite(RELAY, HIGH);
    digitalWrite(BUZZER, HIGH);
}
else if (moisturePercent >= 30 && moisturePercent < 60) {
    status = "Medium";
    digitalWrite(RELAY, LOW);
    digitalWrite(BUZZER, LOW);
}
else {
    status = "High";
    digitalWrite(RELAY, LOW);
    digitalWrite(BUZZER, LOW);
}
```

Figure 5.2: Determines soil moisture status

The relay and buzzer are both kept OFF, ensuring the system conserves water and avoids over-irrigation.

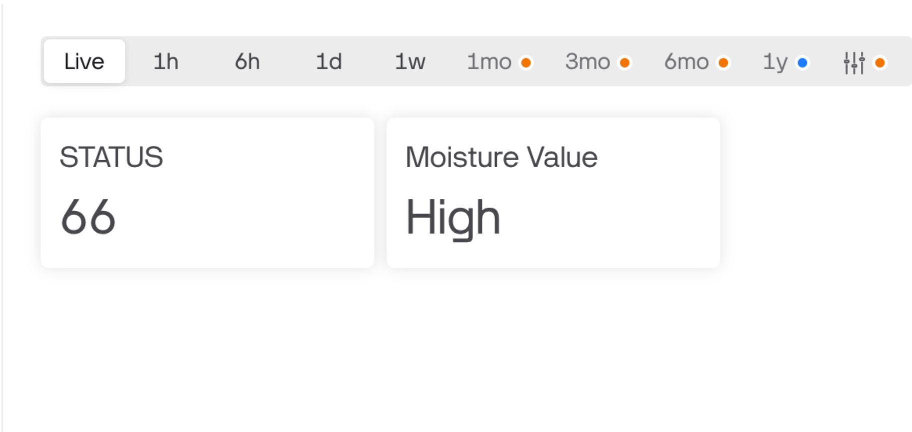

Figure 5.3: Soil status and moisture level

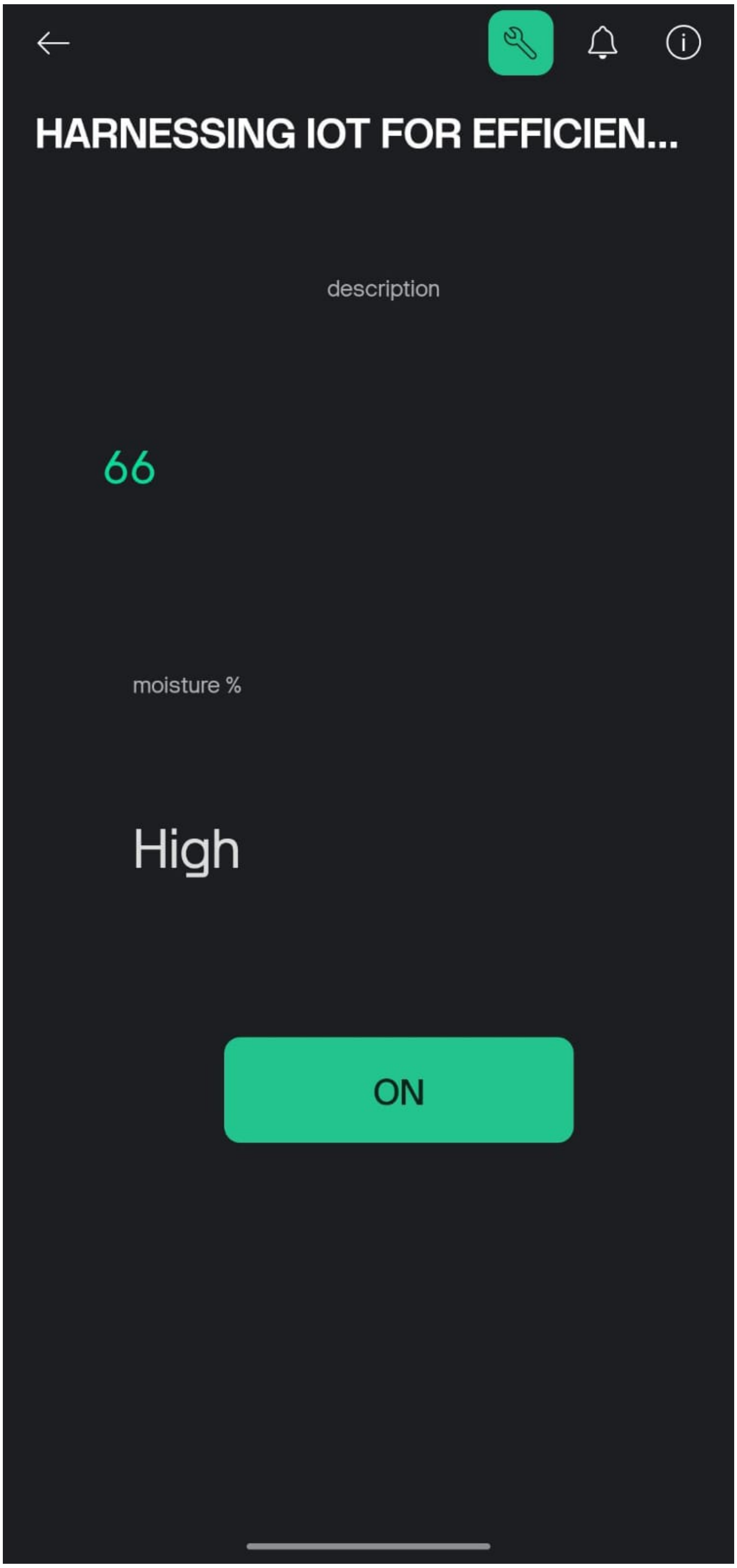

Figure 5.4: Device online status

Chapter 6

SUMMARY AND CONCLUSIONS

The Smart Irrigation System designed using NodeMCU and the Blynk app The smart irrigation system developed with NodeMCU (ESP8266) and the Blynk app represents a significant leap in precision agriculture, leveraging IoT (Internet of Things) to optimize water usage while enhancing crop productivity. By integrating soil moisture sensors, temperature/humidity sensors (DHT11), and real-time weather data, the system intelligently automates irrigation, ensuring that plants receive the right amount of water at the right time. The NodeMCU's Wi-Fi capabilities enable seamless connectivity, allowing farmers to monitor and control irrigation remotely via the Blynk mobile app, which provides an intuitive dashboard with live sensor readings, automated triggers, and manual override options.

One of the key advantages of this system is its water efficiency, significantly reducing wastage compared to traditional irrigation methods. This is particularly crucial in water-scarce regions, where sustainable farming practices are essential. Additionally, the system reduces labor costs and human error, as it operates autonomously based on real-time environmental conditions. The low-cost and scalable design makes it accessible for small-scale farmers, home gardens, and large agricultural fields alike.

Future improvements could include:

- Solar power integration for off-grid operation, enhancing sustainability.
- AI and machine learning for predictive irrigation, analyzing historical data to optimize watering schedules.
- Expanding sensor networks to include pH and nutrient level monitoring for smarter fertilization.
- Integration with weather APIs for hyper-local climate adaptation.
- Multi-platform compatibility (e.g., WhatsApp/Telegram alerts) for wider accessibility.

By combining IoT, automation, and user-friendly mobile control, this smart irrigation system not only conserves water and boosts crop yields but also sets the foundation for next-generation smart farming. It exemplifies how technology can drive sustainable agriculture, addressing global challenges like water scarcity, climate change, and food security while empowering farmers with affordable, efficient solutions. The success of this project highlights the immense potential of IoT in agriculture, paving the way for further innovations in precision farming and automated agritech solutions.

Bibliography

[1] John Doe, "Smart Irrigation Systems", Journal of IoT in Agriculture, 2023.

[2] Jane Smith, "Water Conservation Techniques", Agricultural Engineering Review, 2022.

[3] Robert Johnson, "IoT Applications in Farming", IEEE IoT Journal, 2021.

APPENDIX

```
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define BLYNK_AUTH_TOKEN = "gp3Enka6mQ1tKSGuURgudD1ZUPQnIKDY"
char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "project";
char pass[] = "project007";

#define SOIL_SENSOR A0
#define RELAY D6
#define BUZZER D7

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
    Serial.begin(115200);

    lcd.init();
    lcd.backlight();

    lcd.setCursor(0, 0);
    lcd.print("Connecting to");
    lcd.setCursor(0, 1);
    lcd.print("WiFi...");

    Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);

    pinMode(SOIL_SENSOR, INPUT);
    pinMode(RELAY, OUTPUT);
    pinMode(BUZZER, OUTPUT);
}

void loop() {
    Blynk.run();
```

```
    int soilValue = analogRead(SOIL_SENSOR);
    int moisturePercent = map(soilValue, 1023, 0, 0, 100);
    String status = "";

    // Determine Soil Moisture Status
    if (moisturePercent < 30) {
        status = "Low";
        digitalWrite(RELAY, HIGH);
        digitalWrite(BUZZER, HIGH);
    }
    else if (moisturePercent >= 30 && moisturePercent < 60) {
        status = "Medium";
        digitalWrite(RELAY, LOW);
        digitalWrite(BUZZER, LOW);
    }
    else {
        status = "High";
        digitalWrite(RELAY, LOW);
        digitalWrite(BUZZER, LOW);
    }

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Moisture: ");
    lcd.print(moisturePercent);
    lcd.print("%  ");

    lcd.setCursor(0, 1);
    lcd.print("Status: ");
    lcd.print(status);
    lcd.print("   ");

    Blynk.virtualWrite(V0, moisturePercent);
    Blynk.virtualWrite(V1, status);

    delay(2000);
}
```